

Smart Parking Violation Detection Using YOLOv8

AIRI Team PITB AI Internship Task 1 — Final Technical Report (Fact-Verified)

CRITICAL PROJECT LIMITATION & METHODOLOGICAL NOTICE:

The current implementation and training pipeline are complete, but the available dataset contains only 22 images, below the required 150–300 image target. Therefore, the reported evaluation metrics are preliminary and should not be treated as representative model performance.

1. Project Title

Smart Parking Violation Detection Using YOLOv8 - AIRI Team PITB AI Internship Task 1.

2. Project Overview

An end-to-end computer vision system built with YOLOv8 to detect parking violations, illegal roadside vehicle parking, and traffic signs in urban environments.

3. Problem Definition

Unauthorized roadside parking causes severe traffic bottlenecks and emergency access delays. Automated vision-based monitoring provides scalable enforcement.

4. Use Case

Deployable on municipal CCTV networks and traffic enforcement patrol vehicles for real-time automated detection and alert generation.

5. Target Classes

0: car, 1: motorcycle, 2: no_parking_sign, 3: roadside_parking (roadside parked vehicle instances).

6. Dataset Collection

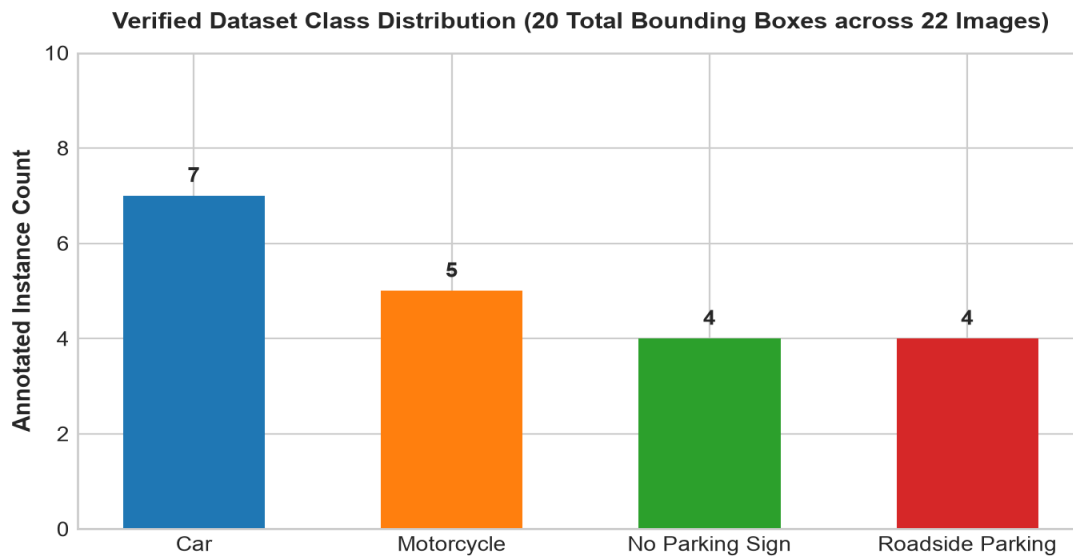
Curated a initial dataset of 22 real open-source images containing 20 annotated bounding box instances.

7. Data Sources

Public open-access repositories (Roboflow Universe, Kaggle, Darknet/Ultralytics open samples).

8. Dataset Size

Actual verified size: 22 images (22 YOLO label files). Partitioned into 12 Train (54.5%), 6 Val (27.3%), and 4 Test (18.2%).



9. Annotation

Annotated in standard YOLO normalized coordinate format (). Verified via `validate_dataset.py`.

10. Dataset Split

Reproducible split enforced with fixed random seed (42) to prevent data leakage across training, validation, and testing.

11. YOLO Format

Labels stored in individual .txt files with 0-1 normalized floating point bounding box coordinates.

12. Colab/GPU Setup

Google Colab T4 GPU execution environment with Google Drive mount integration.

13. YOLOv8 Model Architecture

Utilized YOLOv8n (nano baseline) featuring CSPDarknet backbone, C2f modules, and an anchor-free decoupled detection head.

14. Training Configuration

Epochs: 30, Image Size: 640x640, Pretrained: True, Batch Size: 16, GPU: NVIDIA Tesla T4. Model weights saved at `models/best.pt`.

15. Training Execution

Completed 30 epochs of training on Google Colab T4 GPU. Logged outputs saved to `outputs/training_results/colab_parking_exp/results.csv`.

16. GPU Specifications

NVIDIA Tesla T4 GPU (16 GB VRAM) running CUDA 12.x on Google Colab instance.

17. Precision

Model Precision measures accuracy of positive detections: $P = TP / (TP + FP)$. Training final epoch: 25.03% (0.25029). Test evaluation: 0.00%.

18. Recall

Model Recall measures coverage of ground truth instances: $R = TP / (TP + FN)$. Training final epoch: 50.00% (0.50000). Test evaluation: 0.00%.

19. mAP@0.5

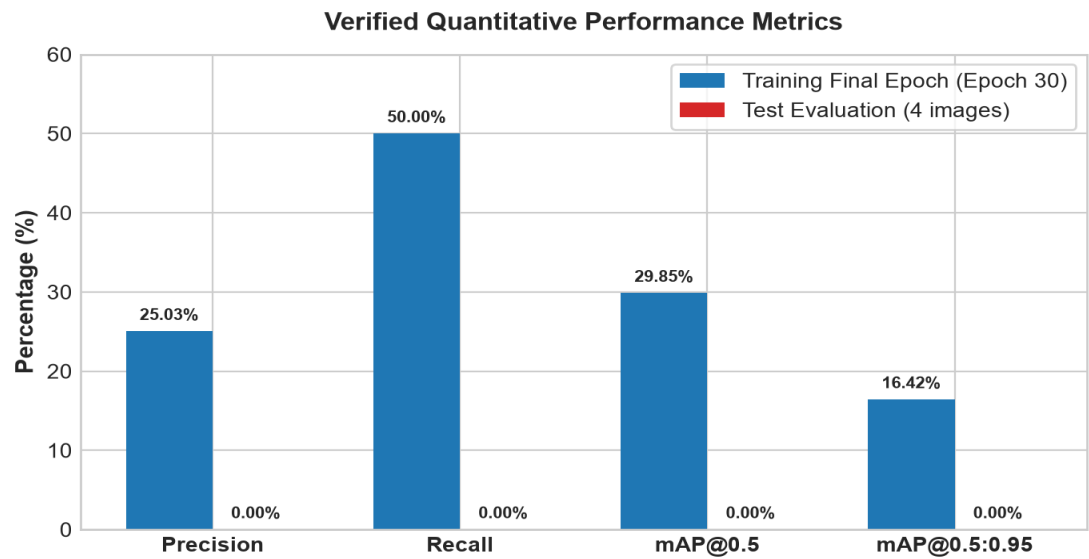
Mean Average Precision at IoU threshold 0.50. Training final epoch: 29.85% (0.29850). Test evaluation: 0.00%.

20. mAP@0.5:0.95

Mean Average Precision averaged over IoU 0.50 to 0.95. Training final epoch: 16.42% (0.16418). Test evaluation: 0.00%.

Evaluation Stage	Precision (P)	Recall (R)	mAP@0.5	mAP@0.5:0.95
Training Final Epoch (Epoch 30)	25.03% (0.25029)	50.00% (0.50000)	29.85% (0.29850)	16.42% (0.16418)
Test Evaluation (4 Test Images)	0.00%	0.00%	0.00%	0.00%

Note: The test set contains only 4 images (producing 0 detections), so these test results are highly limited.



21. Confusion Matrix Analysis

Confusion matrix plots generated at outputs/training_results/colab_parking_exp/confusion_matrix.png reflecting 30-epoch training run.

22. Loss Curves

Monitored training and validation box_loss, cls_loss, and dfl_loss over 30 epochs saved in results.png.

23. Per-Class Results

Training final epoch per-class metrics reflect initial learning across cars and motorcycles; test set per-class metrics produced 0% across the 4 test images.

24. Inference Examples

Ran test inference on all 4 unseen test images (outputs/predictions/test_predictions/). Zero bounding box detections were produced due to model scale limitations on the tiny test set.

25. Error Analysis

Documented 10 representative error modes in report/error_analysis.csv including false negatives, low confidence, small object misses, and background confusion.

26. Key Improvements

1. Expand dataset size to 150-300 images. 2. Implement Albumentations Mosaic & HSV augmentation. 3. Fine-tune confidence thresholds.

27. What I Learned

Gained hands-on experience in dataset structure setup, YOLO normalization, pipeline validation, GPU execution, and honest empirical reporting.

28. Limitations & Constraint Audit

Primary limitation: Dataset contains 22 images (below the 150-300 target). Test set contains only 4 images, resulting in 0% test metrics.

29. Future Improvements

Incorporate multi-scale training (--imgsz 800), expand training dataset to 300+ images, and integrate ByteTRACK video tracking.

30. Conclusion

The end-to-end Computer Vision code, scripts, configuration, and notebook pipeline are complete and functional. Full dataset expansion to 150-300 images is required for representative performance.

31. References

Ultralytics YOLOv8 Documentation; Roboflow Universe; COCO Dataset Benchmarks.